

PRO CLASSROOM

Minor Project 1st report

Submitted in partial fulfillment of the requirement for the award of the degree of

MASTERS OF COMPUTER APPLICATION

SUBMITTED BY

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Introduction

Nowadays, schools and colleges use online tools so students can easily turn in their assignments. However, manually checking for copied work means teachers have to open each PDF and compare assignments one by one, which takes a lot of time and can lead to mistakes.

With our system, we extract text (even from scanned pages) and automatically compare all submitted PDFs. Instructors then receive a clear similarity percentage for each pair of assignments.

This approach saves teachers a significant amount of time and ensures any copied content is spotted promptly.

Project Objective and Goals

Project Goal

Our project goal is Developing a web-based “Classroom” platform with integrated plagiarism detection that automatically analyses student submissions and reports similarity scores, enabling instructors to review and act on potential academic integrity issues.

In our website the teacher will be able to login and create a new classroom in which they will be able post an assignment that will be seen by the students later when they join the classroom by the respective classroom code. After that the student will be able to submit the assignment which will be checked automatically by our plag-detection system.

Worked Planned (so far)

Tech Stack Defined

- **Front-End:** React.js for a responsive upload and review interface.
- **Back-End:** Node.js (Express) for API endpoints, with a Python microservice handling PDF text extraction and similarity comparisons.
- **Database:** MongoDB for flexible storage of user data, submissions, and similarity results.
- **OCR Engine:** Lama Vision model for detecting and extracting both printed and handwritten text from PDFs.
- **Similarity Algorithm:** Python's built-in difflib library for sequence matching, supplemented by TF-IDF vectorization and cosine similarity in the microservice.

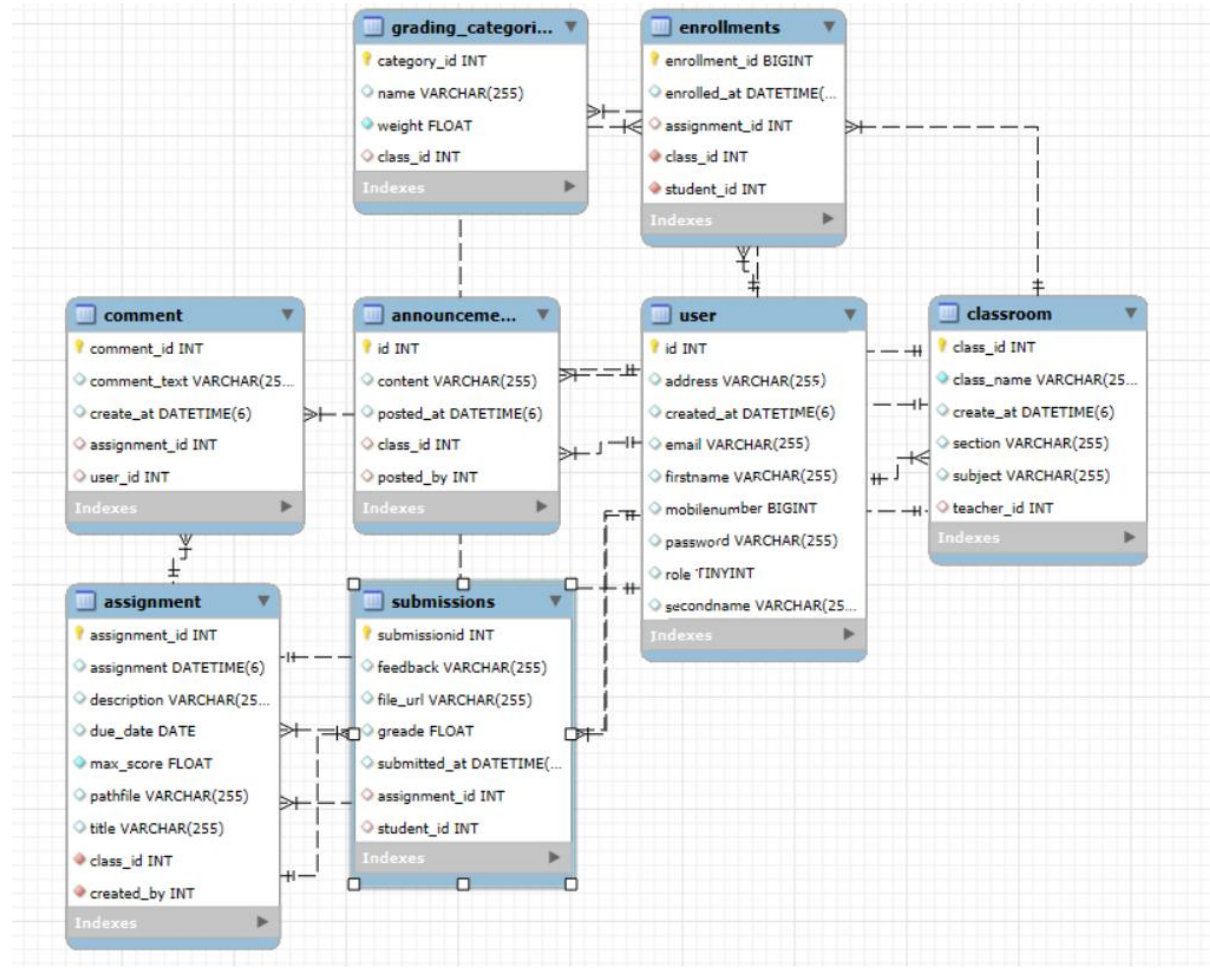
Team & Responsibilities

Front-end : Mohit Kumar and Vishal Singh Rajput

Back-end : Anand Kumar Jha

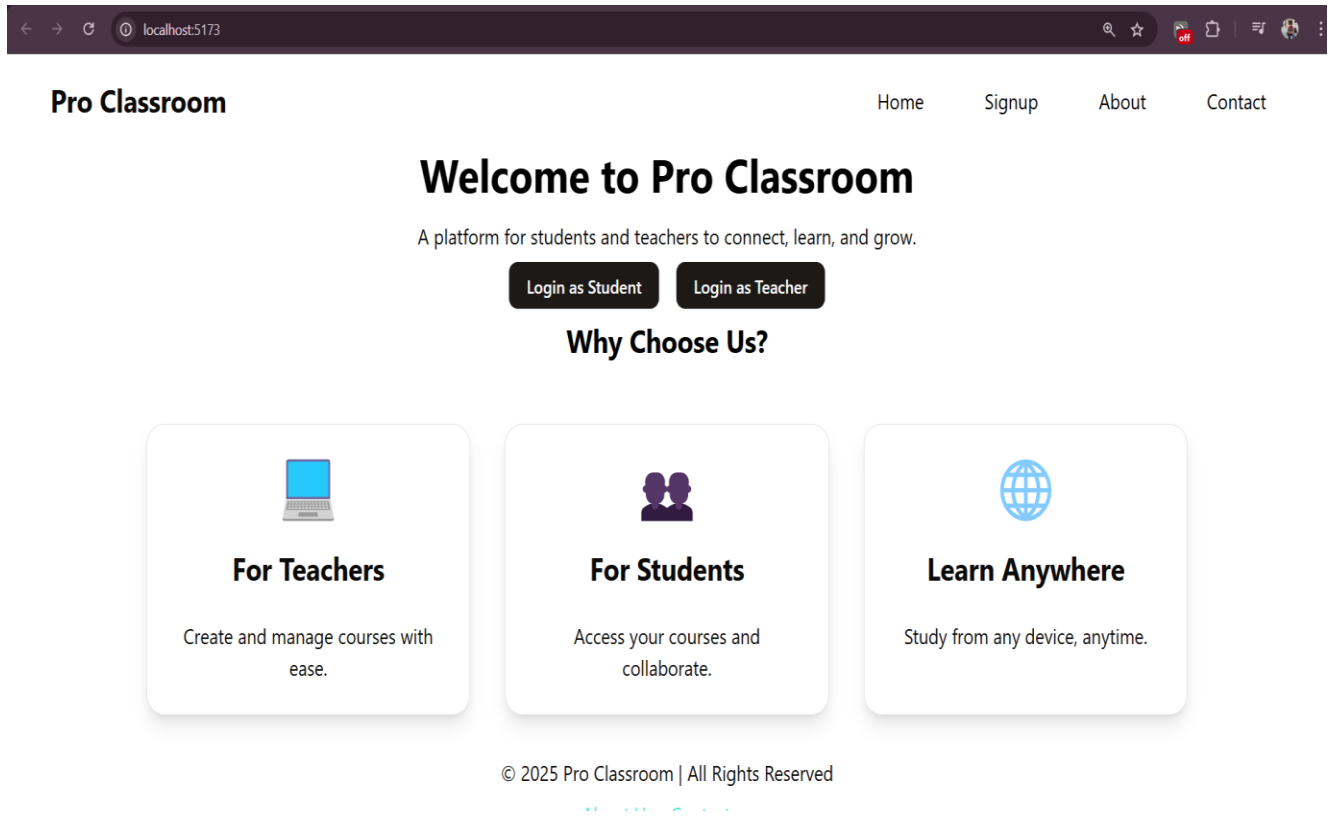
Plag-detection : Keshav

Designed Daigram High Level Daigrams



Work Completed

Designed HomePage



Work in Progress

Development of Backend : Developing of the backend in order to make the website more functional.

Designing Database Schema : Designing the database schema in order to store the information of the teacher and students

Next Steps

- **Implement Authentication Module**

Build signup/login UI and backend

Dependency: Development environment and user schema setup.

- **Assignment Creation & Management**

Develop teacher form for creating/editing assignments (title, description, due date)

Wire in frontend ↔ backend API

- **Student Submission Workflow**

Create upload component (PDF) with client-side validation

Store uploads in local storage.

Conclusion

To this date we have managed to achieve the goal of the project and the project seems to be on time and with the initial requirement finalised we are now looking forward to to achieve our next goal in upcoming four to five weeks.